INCH POUND

MIL-E-46717G
12 April 1989
SUPERSEDING
MIL-E-46717F
31 August 1983

MILITARY SPECIFICATION

ENGINES, GASOLINE: AIR-COOLED, INDUSTRIAL-TYPE, 4-CYCLE, MILITARY-DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers five sizes of light- weight, air-cooled, high-performance, industrial-type, military-design gasoline engines.

1.2 Classification. The engines shall be of the following models as specified (see 6.2):

Model 1A08	- 1.50 net continuous brake horsepower at rated 3,600 rpm.
Model 2A016	- 3 net continuous brake horsepower at rated 3,600 rpm.
Model 4A032	- 6 net continuous brake horsepower at rated 3,600 rpm.
Model 2A042	- 10 net continuous brake horsepower at rated 3,600 rpm.
Model 4A084	- 20 net continuous brake horsepower at rated 3,600 rpm.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2805

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

1.2.1 International standardization agreement code number.

ABCA-Army-STD-162 (see 6.10).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-575	- Bolt, Hexagon and Square.
FF-N-105	- Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
FF-S-92	- Screw Machine: Slotted, Cross-Recessed or Hexagon Head.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-276 - Impregnation of Porous Nonferrous Metal Castings.
MIL-STD-453 - Inspection, Radiographic.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-731 - Quality of Wood Members for Containers and Pallets.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking.
MIL-STD-1226 - Engine, Gasoline: Air-Cooled, 1-1/2 BHP, 4-Cycle, Military Design, Model 1A08 Installation Procedures.
MIL-STD-1227 - Engine, Gasoline: Air-Cooled, 3 BHP, 4-Cycle, Military Design, Model 2A016, Installation Procedures.
MIL-STD-1300 - Engine, Gasoline, Air-Cooled, 6 BHP, Military Design, Model 4A032, Installation Procedures.
MIL-STD-1400 - Engines, Gasoline and Diesel, Methods of Test.
MIL-STD-1401 - Engine, Gasoline: Air-Cooled, 20 BHP, 4-Cycle, Military Design, Model 4A084 Installation Procedures.
MIL-STD-1402 - Engine, Gasoline, Air-Cooled, 10 BHP, 4-Cycle, Military Design, Model 2A042 Installation Procedures.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS16992 - Bolt (Screw), Lag, Hex Head, Gimlet Point.
MS27183 - Washer, Flat-round, Steel, Cadmium Plated, General Purpose.
MS35338 - Washer, Lock-spring, Helical, Regular (Medium) Series.
MS51941 - Nut, Plain, Plate.
MS51967 - Nut, Plain, Hexagon-carbon Steel, Cadmium Plated, UNC-2B.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA9786E3 - Engine, Gasoline, 1-1/2 HP, Military Design, Model 1A08.
TA9787E3 - Engine, Gasoline, 3 HP, Military Design, Model 2A016.
TA13206E0000 - Engine, Gasoline, 6 HP, Military Design, Model 4A032.
TA13206E0500 - Engine, Gasoline, 10 HP, Military Design, Model 2A042.
TA13206E1000 - Engine, Gasoline, 20 HP, Military Design, Model 4A084.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

C13200E9246 - Container, Shipping 1-1/2 and 3 HP.
D13206E0461 - Carton, Shipping, 6 HP.
D13206E0986 - Bracket, Engine, Shipping, 10 and 20 HP.
D13212E7065 - Shipping Rack and End Frame, 10 and 20 HP.
D13226E1229 - Soundness Requirements, Aluminum Alloy Castings, MIL-STD Engines.

USA BELVOIR RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

- Manual for Inspection and Repair of Aluminum Alloy Castings (Dec. 1981).

(Application for copies should be addressed to the USA Belvoir Research, Development, and Engineering Center, Fort Belvoir, VA 22060-5606, ATTN: STRBE-FCS.)

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FGS, Fort Belvoir, VA 22060-5606.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The engines furnished under this specifica- tion shall conform to drawings TA9786E3, TA9787E3, TA13206E0000, TA13206E0500 or TA13206E1000, as applicable, and to the require- ments specified herein. Unless otherwise specified (see 6.2), an air-inlet housing assembly (for the 2A042 and 4A084 only) and a fuel filter shall be furnished. The engines shall be complete with specified accessories, shall have successfully passed the tests specified herein, and shall be ready for immediate operation after deprocessing and servicing. The engines shall be capable of performing as specified herein in any ambient tempera-ture from +125 °F to -25 °F; and at any elevation from sea level and a maximum ambient temperature of 125 °F to 5,000 feet and a maximum ambient temperature of 107 °F.

3.1.1 Drawings. The drawings forming a part of this specification are end product design drawings. Where tolerances prescribed could cumulatively result in improper fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the engines. No deviation from the prescribed dimensions or tolerances and other requirements shown on the drawings is permissible without prior approval of the contracting officer. Any data (shop drawings, layouts, flow sheets,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

processing procedures) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative. The dimensions shown on the drawings apply at temperatures from 68 to 70 °F.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The engines shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the engines may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the engines may be newly fabricated from recovered materials to the maximum extent practicable, provided the engines produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the engines.

3.3.3 Aluminum alloy castings. Aluminum alloy castings shall be specified in the applicable end product design drawings, shall be in accordance with Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings, and as specified herein. Critical castings are identified on the drawings for certain parts. Manufacturers of critical castings shall be qualified in accordance with Drawing 13226E1229.

3.3.3.1 Identification of castings. All castings shall be identified in accordance with Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings. In addition, each casting shall include an external and visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

designation showing the year and month the casting was made.

3.3.3.2 Visual discontinuities (defects). All castings shall be complete (or as permitted in Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings) and shall be free from discontinuities such as cracks, cold shuts, misruns, blowholes, and external shrinks.

3.3.3.3 Machined surfaces. Imperfections in machined surfaces of all castings shall not exceed those permitted in Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings.

3.3.3.3.1 Surface uniformity. Surface uniformity of all castings shall be in accordance with Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings, when subjected to liquid penetrant testing in accordance with MIL-STD-271.

3.3.3.4 Soundness of critical castings. Critical castings shall be radiographically inspected in accordance with MIL-STD-453 and Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings, and shall meet the radiographic standards contained in "Manual for Inspection and Repair of Aluminum Alloy Castings".

3.3.3.5 Impregnation. Sand castings may be impregnated in accordance with MIL-STD-276 provided they are in accordance with all requirements specified herein.

3.3.3.6 Misruns. The total fin area of each engine cylinder, cylinder head or cylinder barrel should be completely filled. However, the total fin area of these castings may be reduced a maximum of 10 percent due to misrun and subsequent blending to minimize sharp or ragged metal edges.

3.4 Fungus and moisture resistance. Unless otherwise specified herein, the engine components and materials shall be treated to resist fungus and moisture as specified on the applicable drawings.

3.5 Performance.

3.5.1 Starting. The engine shall start within 5 minutes and shall operate after 15 minutes of warmup under any of the conditions as specified herein.

3.5.2 Operating. When tested as specified herein, the engines shall operate without malfunction or failure attributable to workmanship or materials. Mating surfaces and connections such as seals, gaskets and fittings (singularly or collectively) shall not leak in excess of globule formation. Globules shall not fall or blow free from the engine under any of the conditions specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

3.5.3 Engine operating temperatures.

3.5.3.1 Models 1A08 and 2A016. The lubricating oil sump temperatures shall not exceed 250 °F, and cylinder head temperature (under the spark plugs) shall not exceed 475 °F at rated-load and 500 °F at wide-open-throttle at any ambient temperature up to 125 °F.

3.5.3.2 Models 4A032, 2A042 and 4A084. The lubricating oil sump temperatures shall not exceed 265 °F and cylinder head temperature (under the spark plugs) shall not exceed 475 °F at rated-load and 500 °F at wide-open throttle at any ambient temperature up to 125 °F, except that during the first 10 hours of engine operation, oil sump temperature shall not exceed 300 °F and cylinder head temperature shall not exceed 500 °F at rated-load and 525 °F at wide-open-throttle.

3.5.4 Oil pressure.

3.5.4.1 Models 4A032, 2A042 and 4A084. Lubricating oil pressure shall be not less than 20 psi at 2,000 rpm nor more than 60 psi at 3,600 rpm under any operating condition specified herein.

3.5.5 Oil consumption. The average specific oil consumption shall not exceed 0.006 lb/bhp/hr.

3.6 Governing. The engines shall have the following governing characteristics at any operating condition specified herein. Two frequency excursions beyond the specified bandwidth are permitted for any 2-hour period for models 1A08 and 2A016 engines and any 4-hour period for engine models 4A032, 2A042 and 4A084. These excursions shall be of a nature clearly indicative of a short-term cyclic transient of a self-correcting nature and not exceeding 2.50 percent beyond the specified bandwidth. For interpretation of governor performance and definition of terms (see appendix, annex I).

Rfn = maximum of 3 percent.
Rnf = maximum of 3 percent.
D = maximum of ±0.50 of 1 percent for models 4A032, 2A042 and 4A084.
D = maximum of ±1 percent for models 1A08 and 2A016.
So = maximum of 5 percent.
Su = maximum of 3 percent.
Tfn = maximum of 6 seconds.
Tnf = maximum of 4 seconds.
Jr = maximum of ±1 percent excursion limits (first article engines and sample production engines only) during any period of 4 hours or less of continuous operation for models 4A032, 2A042 and 4A084 and 2 hours or less of continuous operation for models 1A08 and 2A016.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

Jo = maximum of ±1 percent excursion limits (first article engines and sample production engines only) during any period of 4 hours or less of continuous operation for models 4A032, 2A042 and 4A084 and 2 hours or less of continuous operation for models 1A08 and 2A016.

3.7 Power and speed.

3.7.1 Model 1A08. The model 1A08 engine with all accessories shall be capable of producing not less than 2.5 maximum brake horsepower (bhp) (see 6.6.1). The engine shall also produce not less than 2.10 hp for 500 hours wide-open-throttle at 3,550-3,650 rpm (corrected to standard conditions (see 6.6.2)) or not less than 1.5 continuous bhp (see 6.6.3) for 1,500 hours continuous operation at 3,550-3,650 rpm.

3.7.2 Model 2A016. The model 2A016 engine with all accessories shall be capable of producing not less than 4.8 maximum bhp (see 6.6.1). The engine shall also produce not less than 4.25 hp for 500 hours at wide-open-throttle at 3,550-3,650 rpm (corrected to standard conditions (see 6.6.2)) or not less than 3.0 continuous bhp (see 6.6.3) for 1,500 hours continuous operation at 3,550-3,650 rpm.

3.7.3 Model 4A032. The model 4A032 engine with all accessories shall be capable of producing not less than 10.1 maximum bhp (see 6.6.1). The engine shall also produce not less than 9.00 hp for 500 hours at wide-open-throttle at 3,550-3,650 rpm (corrected to standard conditions (see 6.6.2)), or not less than 6.0 continuous bhp (see 6.6.3) for 1,500 hours continuous operation at 3,550-3,650 rpm.

3.7.4 Model 2A042. The model 2A042 engine with all accessories shall be capable of producing not less than 16.7 maximum bhp (see 6.6.1). The engine shall also produce not less than 15.00 hp for 500 hours at wide-open-throttle at 3,550-3,650 rpm (corrected to standard conditions (see 6.6.2)), or not less than 10 continuous bhp (see 6.6.3) for 1,500 hours continuous operation at 3,550-3,650 rpm.

3.7.5 Model 4A084. The model 4A084 engine with all accessories shall be capable of producing not less than 33.3 maximum bhp (see 6.6.1). The engine shall also produce not less than 29.75 hp for 500 hours at wide-open-throttle at 3,550-3,650 rpm (corrected to standard conditions (see 6.6.2)), or not less than 20 continuous bhp (see 6.6.3) for 1,500 hours continuous operation at 3,550-3,650 rpm.

3.8 Noise limits. The noise produced by model 1A08 and model 2A016 shall be less than 85 dB(A) at MIL-STD-1474, 5.1.2.1.2 locations, and shall be in accordance with other MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5 4, when tested in accordance with 4.7.9. The provisions of MIL-STD-1474, 4 3 and 4.4 shall be provided if and only if: 1, the MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit at the MIL-STD-1474 5.1.2.1.2 locations is obtained from the procuring activity; or 2, the contractor has no control over the noise output due to the requirement to follow level 3 military drawings; in which case the contractor is encouraged to submit engineering change proposals to reduce the noise below 85 dB(A). Noise hazard signs shall be as specified in MIL-STD-1474, 4.3, except the signs will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET." The (specified) distance will be the maximum distance from the equipment that 85 dB(A) is measured when tested in accordance with 4.7.9. The sign shall be readable at the distance indicated on the sign.

3.9 Electromagnetic compatibility. The engine shall meet the electromagnetic interference limits as specified in MIL-STD-461, part 9.

3.10 Treatment and painting. Treatment and painting shall be in accordance with MIL-T-704, type F (for ferrous surfaces) or type G (for non-ferrous surfaces). Manifolds or other high temperature components shall not be painted. Paint shall be as specified in MIL-C-46168, color Green 383. Final assemblies shall be stenciled with the word "CARC", with the month and year of coating in block letters (each letter not in excess of 1 inch), in accordance with MIL-C-46168, in color Lusterless Black.

3.10.1 Externally-exposed surfaces. Unless otherwise specified herein, all externally exposed surfaces of the engine and accessories shall be cleaned, treated, and painted.

3.10.2 Nonvisible surfaces. Unless otherwise specified on the drawings, those aluminum parts that are not visible but are exposed to the cooling air or intake air shall be unpainted and shall be treated in accordance with MIL-C-5541, class 1A, or MIL-A-8625, type and class optional.

3.10.3 Unpainted surfaces. As applicable, the power takeoff shaft, power takeoff pilot, pilot bolt circle, flywheel housing pilot and mounting bolt circle, cooling air outlet housing bolt circles, air restriction indicator, governor ball joints, sliding surfaces of the manual throttle control, electric choke controls, crankcase mounting pads, identification and instruction plates, and decals shall be unpainted and cleaned only.

3.11 Identification marking. The engines, assemblies, subassemblies, and parts shall be identified as specified on the drawings in accordance with MIL-STD-130. Identification plates, when used, shall be in accordance with

MIL-P-514, type I, style as applicable, composition C of type I, grade A, class 1 material.

3.12 Engine adjustments. Each engine offered to the Government for inspection and acceptance shall be adjusted for operation at 1.50, 3, 6, 10, or 20 horsepower as applicable, under the test conditions as specified herein, so

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

the engine will require only minor adjustments for immediate operation at destination after deprocessing and servicing with lubricants.

3.13 Workmanship. Workmanship shall be of a quality to assure delivery of engines which are free of defects resulting from defective material, or from incorrect manufacturing or assembly practices and shall meet the requirements specified herein. Nonqualified or defective components, or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished without prior approval of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.2.1 Vendor item component inspection. Vendor item components and subassemblies shall be inspected prior to installation on engines to determine conformance to the requirements specified herein and on the applicable drawings. These items shall be formed into lots and samples selected in accordance with MIL-STD-105, level II, or quality assurance provisions (QAP) on the drawing, when applicable. Unless otherwise specified herein, acceptable quality level (AQL) shall be 2.5 percent defective. Evidence that the sample selected does not meet the acceptance criteria of MIL-STD-105, or the QAP when applicable, shall be cause for rejection of the lot which the sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

represents. Rejected lots shall be screened and resubmitted for acceptance in accordance with MIL-STD-105. Defective items shall be removed from production and held in a bonded area pending disposition.

4.1.2.2 Disassembly inspection. Engines failing any test shall be disassembled in the presence of a Government representative to the extent necessary to determine cause of failure. Each disassembled part shall be inspected in detail for compliance with the requirements specified herein and referenced drawings in regard to performance, materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected, and the engine shall be rebuilt and retested as necessary. Any major deficiency of a workmanship or material nature occurring during or as a result of tests specified shall be cause for refusal to continue acceptance of engines by the Government until evidence has been provided by the contractor that corrective action has been taken to eliminate the conditions causing the deficiency.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification of casting manufacturers (see 3.3.3).

b. Casting inspection (see 4.3).

c. First article inspection (see 4.4).

d. Quality conformance inspection (see 4.5).

e. Comparison inspection (see 4.8).

f. Inspection of packaging (see 4.9).

4.3 Casting inspection.

4.3.1 Sampling. Sampling of all castings for inspection shall be in accordance with Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings.

4.3.2 Inspection. Sample castings selected in accordance with 4.3.1 shall be inspected for the defects, specified in Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings. In addition, samples of critical casting shall be radiographically inspected as specified in Drawing 13226E1229, Soundness Requirements, Aluminum Alloy Castings.

4.4 First article inspection.

4.4.1 First article examination and short-term tests. Ten first article engines shall be examined and tested by the contractor as specified in 4.6.1 and 4.7.1 through 4.7.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.4.2 First article examination and endurance tests. Five additional first article engines shall be examined and tested by the contractor as specified in 4.6.1 and 4.7.8.

4.4.3 Component examination. One additional first article engine shall be completely disassembled, and those parts where a QAP requirement is included on the applicable drawing shall be examined and measured by the contractor and the Government to determine conformance to QAP requirements. When assembly and subsequent disassembly may alter a dimension, the QAP requirement must be certified by prior inspection. Any deviation from requirements shall be recorded and corrected on all the first article engines and on all engines subsequently produced and accepted by the Government. Upon completion of this examination, parts acceptable to the Government may be used in production engines.

4.4.4 Interchangeability inspection. Ten additional first article engines shall be furnished to the Government for examination in accordance with 4.6.1, and subjected to the tests marked "X" in column 1 of table II to determine conformance to the requirements of this specification. Failure of any test shall be cause for performing the inspection specified in 4.1.2.2. The inspection and tests will be performed by the Government at a site selected by the Government. Acceptance of a first article engine shall not exclude the remaining engines from the quality assurance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of this inspection test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the interchangeability test. Failure of such additional tests shall have the same effect as failure of those tests specified as interchangeability tests.

4.4.5 Inspection failure. Failure of any first article engine to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.4.4, respectively, shall be cause for rejection of the first article engines and shall be cause for refusal by the Government to accept production engines until evidence has been provided by the contractor that corrective action has been taken to eliminate deficiencies. Correction of deficiencies attributable to noncompliance with referenced drawings or workmanship shall be accomplished by the contractor at no cost to the Government on engines previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all engines accepted prior to the completion of first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be selected from engines which have successfully completed the examination specified in 4.5.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.5.1.1 Examination. Each production engine shall be examined for the major and minor defects as specified in 4.6.1.

4.5.2 Tests.

4.5.2.1 Individual. Each production engine shall be subjected to the tests marked "X" in column 2 of table II in the order listed, except that the operating time for the rated load test shall be 15 minutes and the maximum power required shall be not less than 2.1 hp for model 1A08, 4.25 hp for model 2A016, 9.0 hp for model 4A032, 15 hp for model 2A042, and 29.75 hp for model 4A084. During the rated load tests on models 2A042 and 4A084, the battery charging system shall be maintained at not less than 25 volts and 3 amperes while delivering current to a resistive load circuit without a battery. All maximum power values shall be corrected to standard conditions (see 6.6.2). Failure of any test shall be cause for rejection.

4.5.2.2 Samples. Engine lots for sample testing shall be formulated from engines which have successfully passed the individual tests specified in 4.5.2.1. All of the engines comprising a lot shall have components of the same manufacturer. Sampling for testing shall be in accordance with MIL-STD-105. Inspection level S-4 shall apply until such time as a lot sample at level S-4 has been accepted on first submittal. After acceptance of a lot sampled at S-4 level and accepted upon a first submittal, level S-2 shall be used for subsequent lots. Tightened and reduced inspection shall be instituted in accordance with MIL-STD-105 criteria. Samples selected shall be subjected to the tests marked "X" in column 3 of table II in the order listed. AQL of 1.5 percent defective shall apply until such time as a lot sample has been accepted on the first submittal. After acceptance of a lot sampled at an AQL of 1.5 percent defective and accepted upon first submittal, an AQL of 4.0 percent defective shall be used for subsequent lots. The acceptance of all subsequent lots of engines shall be withheld until objective evidence acceptable to the Government has been provided by the contractor that the deficiency or deficiencies that caused rejection has been eliminated and that the correction is applied to all engines to be accepted.

4.6 Inspection procedure.

4.6.1 Examination. Engines shall be examined as specified herein for the defects listed in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

TABLE I. Examination schedule.

Number	Defects	Req. para.	First article	Quality conf.	Insp. comp.
Major					
101.	Engine and ignition system malfunction, with improper adjustment, damaged or defective wiring. Crankshaft end play not within tolerance.	3.1	X	X	X
102.	First article engine requirements not met.	3.2	X	X	X
103.	Components and materials not as specified.	3.3	X	X	X
104.	Material deterioration prevention and control not as specified.	3.3.1	X	X	X
105.	Dissimilar metals not as specified.	3.3.1.1	X	X	X
106.	Identification of materials and finishes not as specified.	3.3.1.2	X	X	X
107.	Recovered materials not as specified.	3.3.2	X	X	X
108.	Aluminum alloy castings not as specified.	3.3.3	X	X	X
109.	Identification of castings not as specified.	3.3.3.1	X	X	X
110.	Visual discontinuities not as specified.	3.3.3.2	X	X	X
111.	Machined surfaces not as specified.	3.3.3.3	X	X	X
112.	Surface uniformity not as specified.	3.3.3.3.1	X	X	X
113.	Soundness of critical castings not as specified.	3.3.3.4	X	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

TABLE I. Examination schedule - Continued.

Number	Defects	Req. para.	First article	Quality conf.	Insp. comp.
Major					
114.	Impregnation of sand castings not as specified.	3.3.3.5	X	X	X
115.	Total fin area of castings not as specified.	3.3.3.6	X	X	X
116.	Fungus and moisture resistance not as specified.	3.4	X	X	X
117.	Power and speed not as specified.	3.7 thru 3.7.5	X	X	X
118.	Hazard signs not as specified.	3.8	X	X	X
119.	Electromagnetic compatibility not as specified.	3.9	X	X	X
120.	Treatment and painting not as specified.	3.10 thru 3.10.3	X	X	X
121.	Identification marking not as specified.	3.11	X	X	X
122.	Engine adjustments not as specified.	3.12	X	X	X
123.	Workmanship not as specified.	3.13	X	X	X

4.6.1.1 Unclassified characteristics. All defects having no bearing on function, safety, interchangeability, or life, but which are considered departures from quality workmanship, will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots or ten lots within a 30-day period will be added to the minor classification of characteristics of the specification with no increase in AQL's.

4.6.2 Tests. Tests shall be in accordance with 4.6.2.1, 4.6.2.2, 4.6.2.3, and table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.6.2.1 Test conditions. The following test conditions shall be maintained during performance evaluation tests.

a. Ambient air temperature shall be between 55 and 125 °F.

b. The exhaust gas back pressure imposed by the contractor's exhaust system during the tests shall be not less than 4 or more than 8 inches of water at maximum power and 3,600 rpm for models 1A08, 2A016 and 4A032. There shall be not less than 10 or more than 20 inches of water at maximum power and 3,600 rpm for the 2A042 and 4A084 models.

c. Fuel shall be a regular grade, commercial type.

d. All maximum-power tests shall be conducted with the engines directly coupled to an electrical cradle-type dynamometer. The $_{WR}^{2}$ value of the dynamometer for models 1A08 and 2A016 engines shall produce not more than 410 pound-inches vibratory torque in the shaft between the engine and the dynamometer. The $_{WR}^{2}$ value of the dynamometer for the model 4A032 engine shall produce not more than 658 pound-inches vibratory torque in the shaft between the engine and the dynamometer. These limiting values for vibrating torque shall be calculated in accordance with MIL-STD-1226 for model 1A08, MIL-STD-1227 for model 2A016, and MIL-STD-1300 for model 4A032. For models 2A042 and 4A084, the $_{WR}^{2}$ value of the dynamometer shall be not greater than 216 pound-inches squared, where W is the weight of the rotating mass in pounds and R is the radius of gyration of the rotating mass in inches. The values for $_{WR}^{2}$ shall be calculated in accordance with MIL-STD-1402 for model 2A042 engine and MIL-STD-1401 for model 4A084 engine. All other tests may be conducted with the engine connected to a suitable power absorption device having the same $_{WR}^{2}$ values as specified above, depending on the engine model being tested. An alternate acceptable method for models 2A042 and 4A084 is to connect the engine to a dynamometer having a $_{WR}^{2}$ value not greater than 1,000 pound-inches squared, through use of a single Morflex model 602 flexible coupling or equal.

e. Fuel shall be supplied to the engine carburetor by the engine fuel pump and not by gravity or other means of external pressurization.

4.6.2.2 Instrumentation. Instrumentation shall be provided for engines to measure the following data and shall be installed in accordance with MIL-STD-1400:

a. Engine load.

b. Engine speed (Texas Instrument Co., Portable Rectiriter, single-channel, model PRRHXF-A16-XT, Specification Drawing 164118, or Gould Recorder, model 2105-2202-005542), or equal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

*c. Fuel consumption.

d. Ambient air temperature.

e. Barometric pressure.

f. Intake air temperature.

g. Humidity.

h. Exhaust back pressure (two sides, models 4A032, 2A042 and 4A084).

*i. Exhaust temperature (two sides, models 4A032, 2A042 and 4A084).

j. Lubricating oil temperature.

*k. Lubricating oil consumption.

**l. Cylinder-head temperature (under all spark plugs).

m. Lubricating oil pressure (models 4A032, 2A042 and 4A084).

n. Battery charging system output (voltage and current) models 2A042 and 4A084.

*Not required for engines tested in accordance with 4.5.2.
**No. 1 cylinder head only for engines tested in accordance with 4.5.2.

All instrumentation shall be calibrated to conform to the limits of accuracy as specified in MIL-STD-1400. A record shall be kept of all inspections and calibrations. The cradle-type dynamometer used for the maximum-power test shall be checked with weights prior to each test.

4.6.2.3 Lubrication. The engine crankcase shall be filled to the operating level with oil in accordance with MIL-L-2104, grade 30, except that engines tested in accordance with 4.5.2 shall be filled to the operating level with oil as specified in MIL-L-21260, grade 30.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.6.2.4 Test schedule.

TABLE II. Test schedule.

First article	Quality conf. inspec. Individual Engine	Quality conf. inspec. Sample Engine	Inspection Comparison	Test	Test paragraph	Req. paragraph
1	2	3	4	5	6	7
X	X	X	X	Casting inspection.	4.3	3.3.3
X	-	-	X	Component examination.	4.4.3	3.3
X	X	X	X	Test conditions.	4.6.2.1	3.5
X	X	X	X	Check test and run-in.	4.7.1	3.5
X	X	-	X	Half-load.	4.7.2	3.5
X	X	-	X	No-load.	4.7.3	3.5
X	*X	X	X	Rated-load.	4.7.4	3.7
X	*X	X	X	Maximum power.	4.7.5	3.7
X	X	X	X	Governor.	4.7.6	3.6
X	-	X	X	Final examination.	4.7.7	Section 3
X	-	-	X	Endurance.	4.7.8	3.7
X	-	-	X	Rated load, 1,500 hours.	4.7.8.1	3.7
X	-	-	X	Wide-open-throttle, 500 hours.	4.7.8.2	3.7
X	-	-	X	Noise level.	4.7.9	3.8
X	-	-	X	Electromagnetic interference.	4.7.10	3.9

*See 4.5.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.7 Test procedure.

4.7.1 Check test and run-in. The engine shall be run-in at the following varying speeds to determine that the engine and all instrumentation are operable:

a. 5 minutes of no electrical load at 2,000 to 2,800 rpm.

b. 10 minutes of no electrical load at 3,500 to 3,700 rpm.

At the end of this 15-minute period of operation or at the end of the complete tests as specified in 4.5.2, the engine cylinder-head nuts shall be retorqued as specified in the applicable drawings. Retorquing shall be accomplished after cooling to room temperature.

4.7.2 Half-load. The engine shall be operated at half rated load (within ±0.1 bhp for models 1A08 and 2A016, and within ±0.25 bhp for models 4A032, 2A042 and 4A084) and 3,600 rpm ±100 rpm for 10 minutes. Any irregularities shall be corrected.

4.7.3 No-load. The engine shall be operated without electrical load at no-load governed speed for a period of not less than 5 minutes. Any irregularities shall be corrected.

4.7.4 Rated-load. The engine shall be operated for 50 hours at rated net continuous horsepower and nominal 3,600 rpm (between 3,590 and 3,610 rpm). The engine governor shall control speed within ±18 rpm for models 4A032, 2A042 and 4A084 and ±36 rpm for models 1A08 and 2A016 at the stabilized speed. During this period, the output of the battery charging system on the 2A042 and 4A084 engines shall be maintained at not less than 25 volts and 3.0 amperes while delivering current to a resistive circuit without a battery. Servicing and adjustment in accordance with table A-I of appendix, annex II, as applicable, is permitted. Complete test data (see 4.6.2.2) shall be recorded at 1-hour intervals, except that barometer, vapor pressure and fuel consumption shall be recorded at intervals of not more than 12 hours. The Texas Instrument Chart, or alternate, shall be run continuously at the rate of 12 inches per hour for the duration of the 50-hour test. Adjustment of the carburetor, governor speed or both shall be permitted only at 8-hour intervals after the data are recorded. Adjustment of governor or throttle linkages shall not be permitted. Inability to start and operate for 50 hours while maintaining rated net continuous horsepower (within ±0.1 bhp for 1A08 and 2A016, and 0.25 bhp for models 4A032, 2A042 and 4A084) at the stabilized speed shall constitute failure of this test. Ineffectual operation of alternator charging system, noncompliance with temperature limits, malfunctioning, or evidence of any unusual condition which may cause damage to the engine shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.7.5 Maximum power. The engine shall be operated for a period of not more than 6 minutes at maximum-power output with wide-open-throttle at nominal 3,600 rpm (between 3,950 and 3,610 rpm). During this test, the output of the battery charging system on models 2A042 and 4A084 shall be maintained at not less than 25 volts and 3.0 amperes while delivering current to a resistive circuit without a battery. Adjustment of the carburetor or breaker points (if any) or both, and timing, if adjustable (in accordance with applicable drawings), shall be permitted, as well as setting the governor so that it will not interfere with the throttle setting. Inability of the engine to complete the test, ineffectual operation of alternator charging system, malfunction, evidence of any unusual operation, or failure to produce not less than the wide-open-throttle power specified in 3.7 shall constitute failure of this test.

4.7.6 Governor. Without further adjustments, the engine shall be checked for governing in accordance with appendix, annex I. During this test, the output of the battery charging system shall be maintained at not less than 25 volts and 3.0 amperes while delivering current to a resistive circuit without a battery. Proof of test shall be a Texas Instrument chart using the schedule shown in table III. Trace or chart analysis shall be in accordance with appendix, annex I. Chart speed shall be 6 inches per minute. The following values of Ts, Tg, and To are applicable:

Ts = 30 seconds
Tg = 10 seconds
To = 2 seconds

Inability of the engine to meet the requirements specified in 3.6 shall constitute failure of this test.

TABLE III. Governor test schedule.

Test	Load 1/
1	Rated-load 3/
2	No-load 2/
3	Half-load 3/
4	Rated-load 3/
5	No-load 2/
6	Rated-load 3/

1/ Load shall be accurate to within ±0.1 bhp for 1A08 and 2A016 engines, and ±0.25 bhp for 4A032, 2A042, and 4A084 engines.
2/ No-load is no electrical load on dynamometer except battery charging system on 2A042 and 4A084 engines.
3/ Record load on chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.7.7 Final examination. Upon completion of tests, the engine shall be dismantled, the components and parts shall be visually inspected, and the items listed in table V shall be dimensionally checked. During disassembly, the breakaway torque value (in a tightening direction) of the threaded fasteners shown in table IV may be used as reference to verify that proper techniques and procedures are being used in assembly. Corrective actions shall be taken if required. A complete description of the condition of the engine, including dimensions, shall be recorded. A report that will include all data recorded during the tests of each engine shall be required. Any part that is not within the allowable dimensional tolerance shall be considered excessively worn and shall be rejected. Any excessively worn part shall be cause for refusal by the Government to continue acceptance of engines until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection. Upon completion of this inspection, parts within dimensions allowed after a 50-hour sample test may be used in production engines. Table V will not apply to engines tested in accordance with 4.7.8 (see appendix, annex II).

TABLE IV. Reference torque values.

Item	Inch/Pounds
Models - 1A08, 2A016	
1. Flywheel.	240
2. Bearing cap nuts.	100
3. Connecting rod dipper.	90
4. Connecting rod bolts.	90
5. Cylinder head nuts.	180
6. Muffler manifold nuts.	60
Model - 4A032	
1. Fan.	440
2. Pulley.	110
3. End cover.	110
4. Cylinder head nuts.	180
5. Connecting rod bolts.	90
6. Center main bolts.	40
7. Exhaust manifold nuts.	60
8. Fan housing nuts.	140
Models - 2A042, 4A084	
1. Cylinder studs to crankcase.	125
2. Crankshaft nut.	840
3. Crankshaft diaphragm bolts.	70

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

TABLE IV. Reference torque values - Continued.

Item		Inch/Pounds
4.	Oil pump cover nuts.	75
5.	Camshaft plate screws.	55
6.	Connecting rod nuts.	180
7.	Cylinder head nuts.	260
8.	Rocker shaft nuts.	55
9.	Rocker box cover screws.	20
10.	Intake manifold screw.	50
11.	Oil pan assembly screw.	40
12.	Drain plugs oil pan.	140
13.	Oil pan to crankcase screws.	50
14.	Front gear cover screws.	70
15.	Starter bolt, nuts.	190
16.	Engine mount screws.	130
17.	Spark plug.	275
18.	Fan-flywheel bolt.	950
19.	Fan housing bolts.	280
20.	Pulley nut.	950
21.	Fuel pump to crankcase screws.	55
22.	Magneto to crankcase bolt, nuts.	180
23.	Exhaust manifold nuts.	70
24.	Oil filter cover bolt.	70
25.	Governor to crankcase screws.	55
26.	Alternator to cover screws.	20
27.	Magneto adapter (nut).	350
28.	Center main bearing nuts.	80

TABLE V. Dimensional inspection.
(Allowed after 50-hour sample test.)

Item		Models	
		1A08	2A016
1.	Ignition timing at 3,600 rpm (if adjustable).	25 - 32° BTDC	25 - 32° BTDC
2.	Valve lash (cold). Intake & Exhaust.	0.006 - 0.010	0.006 - 0.010
3.	Rotor clears poles.	0.002 - 0.016	0.002 - 0.016
4.	Camshaft endplay.	0.002 - 0.017	0.002 - 0.015
5.	Camshaft axle dia.	0.3710 - 0.3733	0.3710 - 0.3733
6.	Crankshaft endplay.	0.003	0.003
7.	Rod side clearance.	0.020 - 0.036	0.020 - 0.036

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

TABLE V. Dimensional inspection - Continued.
(Allowed after 50-hour sample test.)

Item		Models	
		1A08	2A016
8.	Rod bore dia.	0.8789 - 0.8810	0.8789 - 0.8810
9.	Piston pin dia.	0.4185 - 0.4195	0.4185 - 0.4195
10.	Crankshaft out of round.	0.001	0.001
11.	Crank pin dia.	0.8735 - 0.8750	0.8735 - 0.8750
12.	Cam lobe wear.	1.3237 - 1.3287	1.3237 - 1.3287
13.	Rod bearing clearance.	0.0039 - 0.0075	0.0039 - 0.0075
		Model 4A032	
1.	Ignition timing at 3,600 rpm (if adjustable).	25 - 32° BTDC	
2.	Valve lash (cold)	0.006 - 0.010	
3.	Crank-C.M., side play.	0.002 - 0.010	
4.	Crank pin diameter.	0.8735 - 0.8750	
5.	Crank-C.M., bearing dia.	1.2485 - 1.250	
6.	Crank-PTO end bearing dia.	1.2485 - 1.250	
7.	Crank-pulley end bearing dia.	0.8735 - 0.8750	
8.	Cam lobe, int. and exh.	1.0566 - 1.0616	
9.	Cam-center, bearing dia.	1.3725 - 1.3740	
10.	Cam-PTO end bearing dia.	0.747 - 0.749	
11.	Cam-pulley end.	0.747 - 0.749	
12.	Piston pin dia.	0.4185 - 0.4195	
13.	Rod bore dia.	0.8789 - 0.8810	
14.	Rod bearing clearance.	0.0039 - 0.0075	
15.	Rod side clearance.	0.020 - 0.036	
		Models 2A042, 4A084	
1.	Ignition timing at 3,600 rpm (if adjustable).	25 - 32° BTDC	
2.	Oil pump gears.	0.499 - 0.500	
3.	Crank-cam gear backlash.	0.004 - 0.008	
4.	Crank end play 10 HP.	0.0015 - 0.0055	
5.	Crank end play 20 HP.	0.0025 - 0.0065	
6.	Crank pin "A" dia.	1.4973 - 1.4980	
7.	Crank pin "A" dia. (1 to "A").	1.4973 - 1.4980	
8.	Crank center main bearing O.D.	1.6254 - 1.6260	
9.	Crank rear bearing O.D.	1.6253 - 1.6260	
10.	Crank front bearing O.D.	1.6253 - 1.6260	
11.	Cam lobe intake.	1.3682 - 1.3731	
12.	Cam lobe exhaust.	1.3682 - 1.3731	
13.	Cam center bearing O.D	1.7187 - 1.7195	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

TABLE V. Dimensional inspection - Continued.
(Allowed after 50-hour sample test.)

Item		Models 2A042, 4A084
14.	Cam rear bering O.D.	0.8738 - 0.8745
15.	Piston pin dia.	0.8591 - 0.8593
16.	Rod bore dia. "A".	1.4984 - 1.5009
17.	Rod bore dia. "B" (1 to"A").	1.4984 - 1.5009
18.	Rod bearing clearance.	0.0004 - 0.0034
19.	Rod side clearance.	0.006 - 0.014
20.	Fuel pump eccentric O.D.	1.307 - 1.317
21.	Center main bearing I.D.	1.6264 - 1.6280
22.	Front diaphragm I.D.	1.6275 - 1.6283
23.	Rocker arm I.D. intake.	0.6253 - 0.6263
24.	Rocker arm I.D. exhaust.	0.6253 - 0.6263
25.	Rocker shaft O.D. intake.	0.6244 - 0.6248
26.	Rocker shaft O.D. exhaust.	0.6244 - 0.6248
27.	Push rod length O/A.	10.199 - 10.213
28.	Cylinder bore size "A" top.	2.999 - 3.0015
29.	Cylinder bore size "B" bottom (11 to "A").	+0.001 taper
30.	Cylinder bore size "C" top (1 to "A").	2.999 - 3.0015
31.	Cylinder bore size "D" bottom (11 to "C").	+0.001 taper
32.	Crankcase center main bearing I.D.	4.8015 - 4.8022
33.	Crankcase rear bearing I.D.	1.6275 - 1.6283
34.	Crankcase cam bearing I.D.C.	1.7215 - 1.7223
35.	Crankcase cam bearing I.D.R.	0.8765 - 0.8771
36.	Crankcase tappet I.D. intake.	0.8760 - 0.8771
37.	Crankcase tappet I.D. exhaust.	0.8760 - 0.8771
38.	Tappet O.D. intake.	0.8740 - 0.8745
39.	Tappet O.D. exhaust.	0.8740 - 0.8745

4.7.8 Endurance. Endurance tests shall be conducted on engines as follows, and shall be run concurrently with other tests specified in 4.4. Inability of the engines to meet the applicable requirements specified in 3.5, 3.6, or 3.7, or servicing, adjustment or part replacement, except for those listed in table A-I of annex II shall constitute failure of this test.

4.7.8.1 Rated-load, 1,500 hours. Rated-load tests of 1,500 hours shall be conducted on each of three engines in accordance with the endurance test procedure (see appendix, annex II). Servicing and adjustment shall be in accordance with appendix, annex II, table A-I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.7.8.2 Wide-open-throttle, 500-hours. Wide-open-throttle tests of 500 hours shall be conducted on two engines in accordance with the endurance test procedure (appendix, annex II). Servicing and adjustment shall be in accordance with appendix, annex II, table A-I.

4.7.9 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum noise levels shall be measured when equipment is operating under full load, or other operating conditions resulting in the maximum noise level output. A MIL-STD-1474, 5.1.2.1.4 contour shall be taken at not fewer than 12 equal (horizontal) arc increments. Additonally, the noise level at the MIL-STD-1474, 5.1.2.1.2 locations shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.7.10 Electromagnetic interference. The engine shall be tested in accordance with the test methods in MIL-STD-461, part 9.

4.8 Comparison inspection. The Government may select engines at any time during the contract production period and subject these engines to the examination specified in 4.6.1, and to the test specified in column 4 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government, and will not include the previously inspected first article engines. In addition to any test specified as part of the comparison inspection, the Government reserves the right to conduct any and all other tests contained in this specification as part of the comparison inspection and failure of such additional tests shall have the same effect as failure of those tests specified as comparison inspection.

4.8.1 Inspection failure. Failure of an inspection comparison engine to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison engine(s) and shall be cause for refusal by the Government to continue acceptance of production engines until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on engines previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all engines accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

4.9 Inspection of packaging.

4.9.1 First article pack inspection.

4.9.1.1 Examination. The first article pack shall be examined for the defects specified in 4.9.2.2. Presence of one or more defect shall be cause for rejection.

4.9.2 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.9.2.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.9.2.2 Examination. Samples selected in accordance with 4.9.2.1 shall be examined for the following defects. AQL shall be 1.0 percent defective.

124.	Engines not preserved in accordance with the referenced document and specific requirements.
125.	Shipping containers not as specified for level A.
126.	Fuel filter and rope not packed as specified for level A or C.
127.	Engines not packed as specified for level A.
128.	Engines not packed in accordance with the referenced document or drawings for level C.
129.	Engines shipped on racks not shrouded as specified for level C.
130.	Depreservation guide not prepared and attached for level A or C.
131.	Marking missing, illegible, incorrect or incomplete for level A or C.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the timeframe specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article model engine or a production engine. If the first article model is utilized, any packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model engine and production engine.

5.2 Preservation. Each engine shall be preserved in accordance with the level A preservation and packaging requirements of MIL-E-10062 as specified for type IV, method I; except governor ball joints shall not be preserved and the fuel filter and starter rope shall be contained together in a heat-sealed, waterproof bag. The filter shall be cushioned within the bag to prevent damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. Each engine, preserved as specified in 5.2, shall be packed as follows:

a. 1.5 hp and 3 hp in accordance with figure 1.

b. 6 hp in accordance with figure 2.

c. 10 hp and 20 hp in accordance with figure 3.

The fuel filter and starter rope, preserved as specified in 5.2, shall be placed and secured within the box, with the applicable engine, in a manner to prevent movement or damage.

5.3.2 Level C.

5.3.2.1 1.5-, 3-, and 6-hp engines. Each engine, preserved as specified in 5.2, shall be packed in accordance with MIL-STD-1190. As an alternate, the following methods of packing have been found to be acceptable and may be utilized at the option of the contractor:

a. 1.5 hp and 3 hp in accordance with Drawing C13200E9246, except the box shall be class domestic.

b. 6 hp in accordance with Drawing D13206E0461, except the box shall be class domestic, variety DW, grade 350.

5.3.2.1.1 Fuel filter and starter rope. The fuel filter and starter rope shall be packed as specified in 5.3.1.

5.3.2.2 10- and 20-hp engines. Unless otherwise specified, each engine, preserved as specified in 5.2, shall be packed in accordance with MIL-STD-1190. When specified (see 6.2), engines shall be shipped on racks as specified in 5.3.2.2.1.

5.3.2.2.1 Engines shipped on racks. The engines shall be secured to the racks by bolting and using 0.75-inch wood shims under the permanent mounting brackets of the engine. The racks shall be Government-furnished (see 6.2) and shall be in accordance with Drawing D13212E7065. The engine shipping bracket shall be in accordance with Drawing D13206E0986. The fuel filter and starter rope pre-served as specified in 5.2 shall be secured to the engine. Engines mounted on the racks shall be shrouded with a minimum 4-mil-thick polyethylene bag to completely cover each engine. The bag shall be secured to the engines by any suitable means.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

5.4 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide", and secured on the engine. Unless otherwise specified (see 6.2 and 6.9), DA Form 1397 shall be prepared.

5.5 Marking. In addition to any special or identification markings required by the contract or order (see 6.2), marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Engines covered by this specification are intended to power military, industrial-type equipment.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Engine model required (see 1.2).

d. Date of issue of DODISS applicable and exceptions thereto (see 2.1.1).

e. When a fuel filter and an inlet housing assembly are not required (see 3.1).

f. When a first article is not required (see 3.2).

g. Timeframe required for submission of first article engines (see 3.2). The contracting officer should require that all testing of first article engines required by 3.2 will be completed within 120 days after submission of the engines.

h. Timeframe required for submission of the first article pack (see 5.1).

i. Level of packing required (see 5.3).

j. When 10- or 20-hp engines are to be shipped on Government-furnished racks (see 5.3.2.2 and 5.3.2.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

k. When other than DA Form 1397 is to be used (see 5.4 and 6.9).

l. Any special marking (see 5.5).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to so in the solicitation.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack and mark the engines in accordance with this specification.

6.5 Manual for inspection and repair of aluminum alloy castings. The contracting officer should arrange to furnish the contractor one copy of the manual titled "Manual for Inspection and Repair of Aluminum Alloy Castings", which contains the radio-graphic standards for aluminum alloy castings.

6.6 Definitions. The following definitions apply to the text of this specification.

6.6.1 Maximum brake horsepower. The maximum brake horsepower is that power which the engine will produce at wide-open-throttle at any speed within the operating speed range for periods up to 5 minutes of continuous operation.

6.6.2 Standard operating conditions. Standard operating conditions are 29.92 inches of mercury barometric pressure and 60 °F air temperature.

6.6.3 Continuous brake horsepower. Continuous brake horsepower is that power which the engine will produce at any speed within the operating range for periods of 1 hour or more of continuous operation.

6.6.4 Quality assurance provisions for drawings. A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

6.7 Quality assurance provisions for records. The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken, when applicable.

6.8 Racks. When required, the contracting officer shall arrange for furnishing the shipping racks (see 5.3.2.2 and 5.3.2.2.1).

6.9 Depreservation guide. The contracting officer should arrange to furnish DA Form 1397 when requested by the contractor (see 5.4 and 6.2).

6.10 International standardization agreement. Certain provisions of this specification (see 1.2.1) are the subject of international standardization agreement ABCA-Army-STD-162. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.11 Subject term (key word) listing.

Engine, air-cooled
Engine, gasoline
Engine, high-performance
Engine, industrial-type
Engine, 4-cycle

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

Preparing activity:
Army - ME

Review activity:
Army - SM

Project 2805-0542

User activities:
Army - AT, AV, GL,
Navy - MC

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

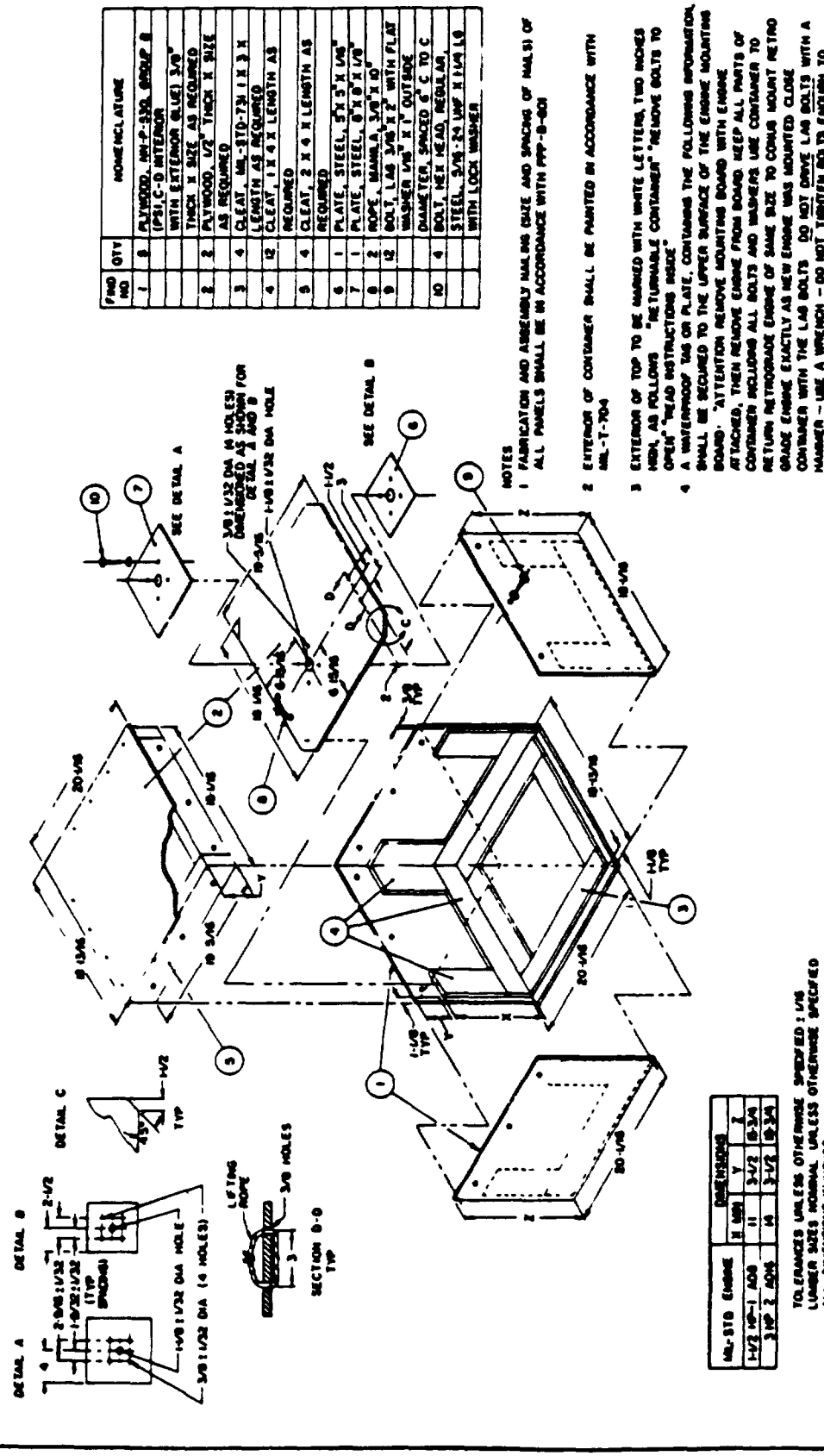


FIGURE 1. Shipping container, 1-1/2 & 3 h.p. military standard engines, fabrication & assembly.

X-1864B

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

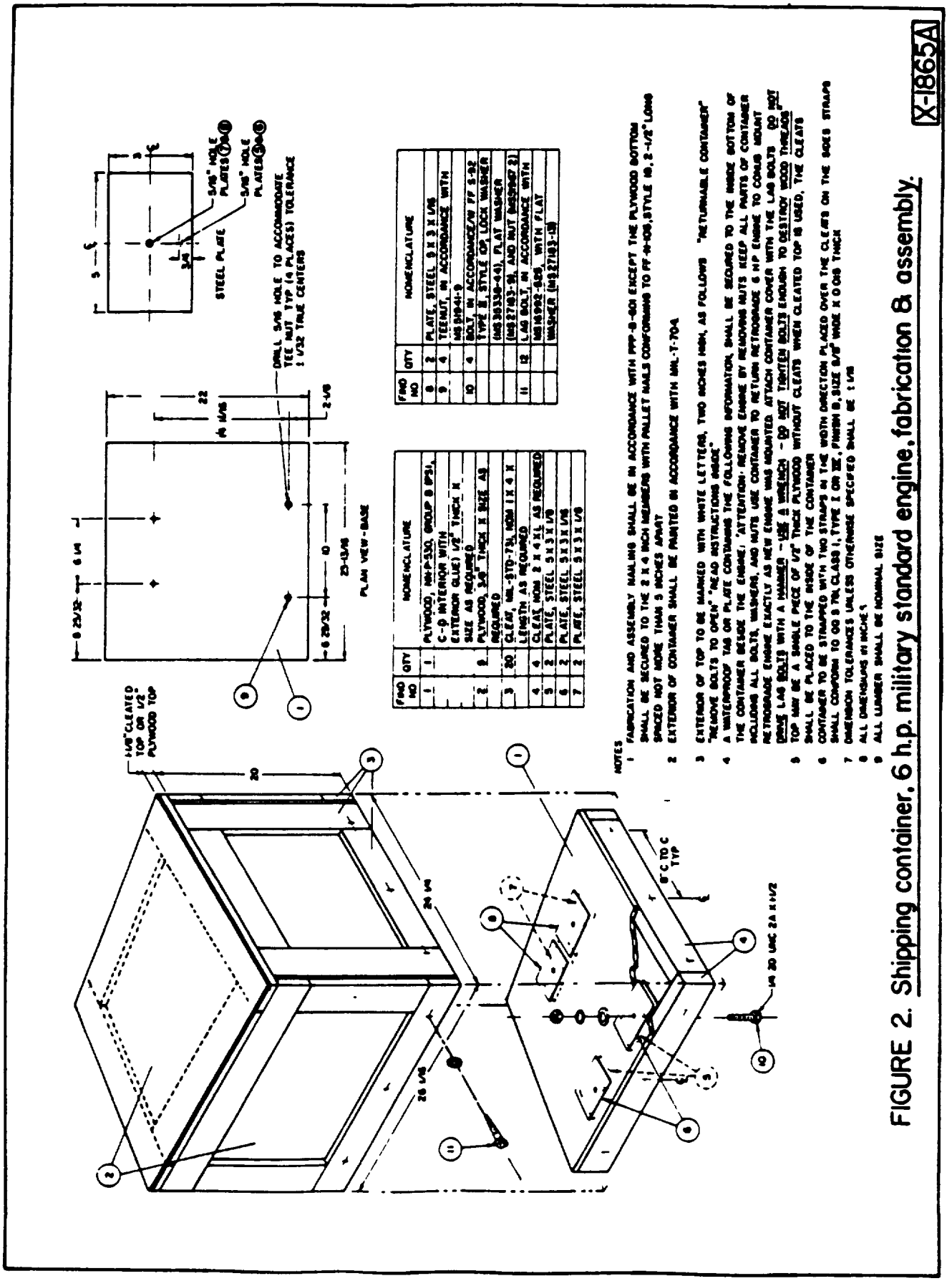
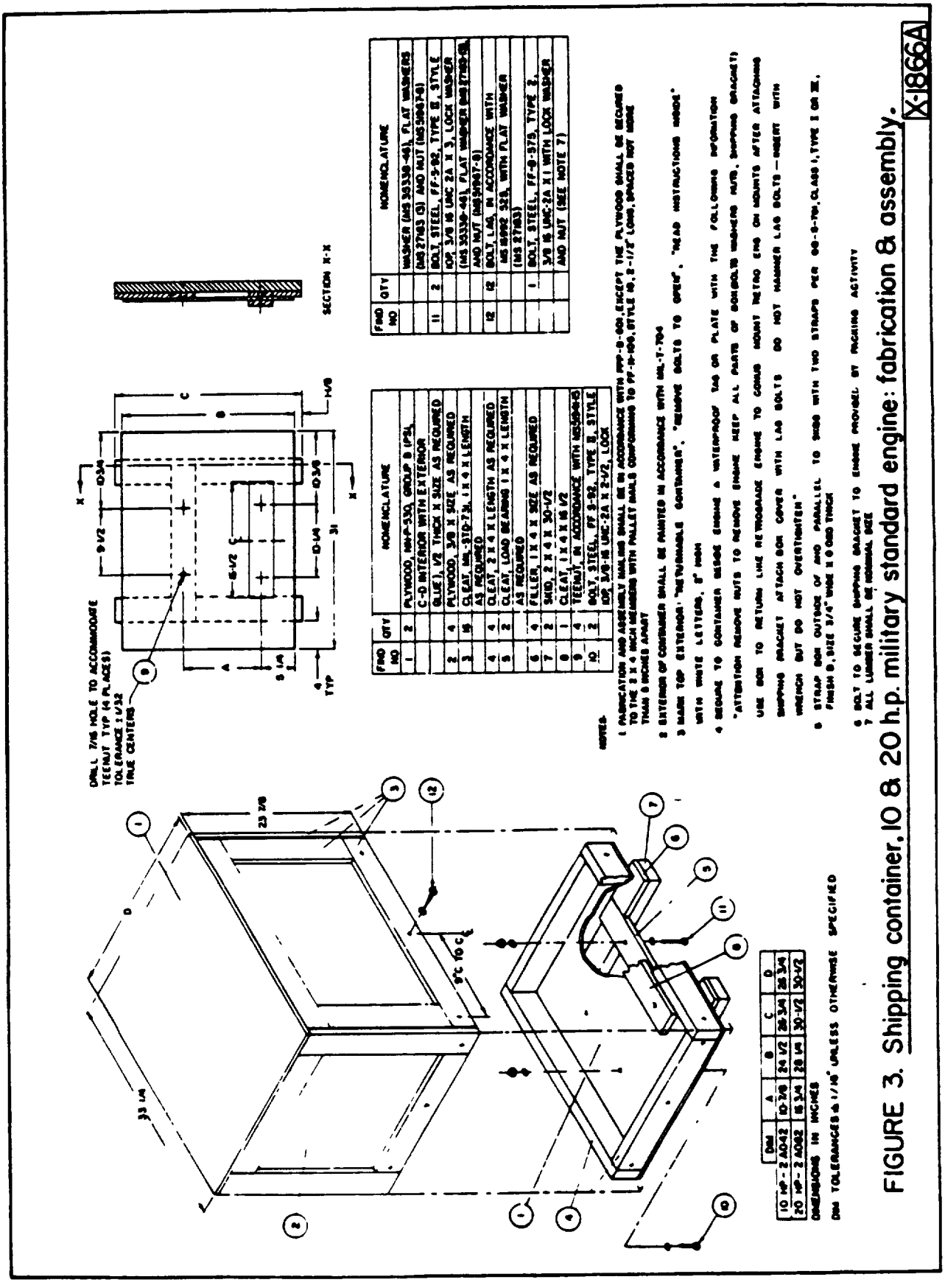


FIGURE 2. Shipping container, 6 h.p. military standard engine, fabrication & assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

APPENDIX

ENGINES, GASOLINE: AIR COOLED,

INDUSTRIAL-TYPE, 4 CYCLE, MILITARY DESIGN

10. SCOPE

10.1 Scope. This appendix details the procedures for required tests, test conditions, and applicable atmospheric standards for engines covered by this specification. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Federal specification.

VV-G-1690 - Gasoline, Automotive, Leaded or Unleaded.

20.2 Other publications.

U.S. Department of Commerce - Weather Bureau Psychrometric Tables.
Publication 235

30. SUBMISSION

30.1 Tests.

30.1.1 Engine governor test. The procedure for engine governing trace analysis, engine governing regulation, transient response, and hysteresis analysis shall be in accordance with annex I of this appendix.

30.1.2 Endurance testing. The endurance test procedures for models 1A08, 2A016, 4A032, 2A042, and 4A084 military design engines shall be in accordance with annex II of this appendix.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

ANNEX I

PROCEDURE FOR ENGINE GOVERNING TRACE ANALYSIS, ENGINE GOVERNING REGULATION, TRANSIENT RESPONSE AND HYSTERESIS ANALYSIS

1.0 SCOPE

1.1 Scope. The purpose of these tests is to determine the procedure for engine governing trace analysis, engine governing regulation, transient response, and hysteresis analysis.

1.2 Definition of terms. The following terms are defined for understanding the engine governor test procedure:

a. A = Actual instrument trace of function.

b. B = Observed speedband (two lines parallel to the axis of chart movement, one each passing through the points of maximum and minimum trace excursion, respectively, during the sample time, Ts).

c. C = Median of observed speedband.

d. Cf1 = Rated-load median, first check.

e. Cf2 = Rated-load median, second check.

f. Cf3 = Rated-load maximum excursion during drift time.

g. Cn1 = No-load median, first check.

h. Cn2 = No-load median, second check.

i. Cn3 = No-load maximum excursion during drift time.

j. D = Specification required steady state speedband.

k. E = Point at which trace leaves rated-load prescribed speedband under condition of decrease in load.

l. F = Point at which trace enters and remains within prescribed no-load speedband.

m. G = Point at which trace leaves no-load prescribed speedband under condition of increase in load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

n. H = Point at which trace enters and remains within rated-load prescribed speedband under conditions of increase in load.

o. Jo = Drift at no-load (maximum excursion).

p. Jr = Drift at rated-load (maximum excursion).

q. K = Hysteresis (inability of a governor to return to the same speed after going through a load change).

r. R = Speed regulation between any two loads.

s. Rfn = Regulation, rated-load to no-load.

t. Rnf = Regulation, no-load to rated-load.

u. S = Speed surge after load change.

v. So = Overspeed surge.

w. Su = Underspeed surge.

x. Tdr = Drift time sample at rated-load.

y. Tdo = Drift time sample at no-load.

z. Tg = Base time for determining position of test sample after a load change.

aa. Tfn = Recovery time, rated-load to no-load.

bb. Tnf = Recovery time, no-load to rated-load.

cc. To = Base time for ensuring that the position of test sample, Ts, is taken prior to load change.

dd. Ts = Test time sample required by specifications for determining stability.

1.3 Procedure. The following procedure shall be used for engine governor testing:

a. Set rated-load speed as close to the rated speed as possible. Mark the left end of the first Ts (figure A-2) reading from right to left. After a specified time, To (figure A-2), reduce the load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

b. Establish that the first rated-load governing trace (figure A-2) was stable within requirements during time, Ts, in accordance with procedure shown on Stability Trace Analysis Sheet (figure A-1).

c. Repeat step 2 for the adjacent no-load curve, allowing 10 seconds for Tg, then taking test time sample, Ts.

d. From point E and point F, determine recovery time, Tfn, by reading distance from chart and converting to time.

e. From lines Cf1 to Cn1, determine regulation (Rfn).

f. From line Cn1, determine overspeed surge (So).

g. The second rated-load trace is checked by the same method as the first one, allowing 10 seconds for Tg prior to Ts.

(1) Repeat 1.3(b) to establish rated-load stability during time, Ts.

(2) From point G to point H, determine recovery time (Tnf).

(3) From Cn1 to Cf2, determine regulation (Rnf).

(4) From Cf2, determine underspeed surge (Su).

h. Compare all values with specification requirements. If not otherwise indicated, specification requirements are met when the calculated or observed values are less than the specification values. The analysis process is repeated for each subsequent load change.

1.4 Engine governing sample calculations. Assumed values for model 4A032 have been used in sample calculations.

a. No calculations are required to establish To. The length of time, Ts, on the chart is calculated in accordance with the notes on figure A-1.

b. Using the rated-load values of stability as used in the stability analysis (figure A-1), the first rated-load trace on figure A-2 is calculated in the following manner:

(1) Line 1 = 59.4.

(2) Line 2 = 59.1.

(3) Line Cf1 = (59.4 + 59.1) divided by 2 = 59.25.

(4) Stability required (see 3.6) by specification = $\pm$0.50 of 1% = $\pm$0.3 cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

(5) Line 4 = 59.25 + 0.3 = 59.55.

(6) Line 5 = 59.25 - 0.3 = 58.95.

The governor is stable within specification at rated-load because of the stability analysis of figure A-1.

c. On the adjacent no-load trace of figure A-2:

(1) Line 1 = 60.90.

(2) Line 2 = 60.55.

(3) Line Cn1 = (60.90 + 60.55) divided by 2 = 60.72 cycle.

(4) Specification = 0.50 of 1% = (60 X .01) divided by 2 = $\pm$ 0.3 cycle.

(5) Line 4 = 60.72 + 0.3 = 61.02 cycles.

(6) Line 5 = 60.72 - 0.3 = 60.42 cycles.

The governor is stable within specification at no-load.

d. Tfn (figure A-2) = 1.1 inches (chart speed = 12 in. min.)
Tfn = 1.1 divided by 12 = 0.092 minute
= 0.092 x 60 = 5.5 seconds

Specification requirement for Tfn = maximum of 6 seconds. Rated-load to no-load recovery time is satisfactory because Tfn (5.5 seconds) is less than the maximum allowed of 6 seconds.

e. Regulation minus rated to no-load (Rfn).
Rfn = Cn1 - Cf1
= 60.72 - 59.25
= 1.47 cycles
= 1.47 divided by 60 = 2.5%

Specification requirement for Rfn = maximum of 3 percent. The Rfn (2.5 percent) is within specification, since it is less than the maximum allowed of 3 percent.

f. Overspeed surge minus the maximum overspeed surge as read from the chart is 63.5 cycles
So = maximum surge - Cn1
= 63.5 - 60.92
= 2.58
= 2.58 divided by 60 = 4.3%

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

Specification requirement for So is a maximum of 5 percent. The So (4.3 percent) is satisfactory, since it is less than the maximum allowed of 5 percent.

g. The load is removed and a second no-load curve results with:

(1) Cf2 = 59.3 (Stability checked as before)

(2) Recovery time Tnf:
Tnf = 0.78 inches (chart speed = 12 in. min.)
Tnf = 0.78 divided by 12 = 0.065 minute
Tnf = 0.065 X 60 = 3.90 seconds

Specification requirement for Tnf = maximum of 4 seconds. Since Tnf (3.90 seconds) is less than the maximum allowed of 4 seconds, the recovery time from no-load to rated-load is satisfactory.

(3) Regulation (no-load to rated-load) = Rnf
Rnf = Cn1 - Cf2
= 60.72 - 59.3
= 1.42 cycles

Specification requirement = maximum of 3 percent = 0.03 X 60 = 1.80 cycles. Since Rnf (1.42) is less than maximum allowed (1.80), the no-load to rated-load regulation is satisfactory; in percent, the value is:

= Regulation (cycles) divided by rated frequency (cycles)
= 1.42 divided by 60 = 2.37%

(4) Underspeed surge minus the maximum underspeed surge (Su) as read from the chart is 57.8.
Su = Cf2 - maximum surge
= 59.3 - 57.8
= 1.5 cycles

Specification requirement = maximum of 3 percent = 0.03 X 60 = 1.8 cycles. Since Su (1.5 cycles) is less than the maximum allowed (1.8 cycles), the underspeed surge (Su) is satisfactory; expressed as a percent, the value is:

= Su (cycles) divided by rated frequency (cycles)
= 1.5 divided by 60 = 2.5%

1.5 Engine governor drift analysis (figure A-3).

a. In accordance with the procedure on figure A-1, establish that the no-load governing is stable within requirements during test time sample,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

Ts. Without any further adjustments to governor or engine, the load is then changed to rated-load after To.

b. Allow 10 seconds for Tg and mark off test sample, Ts. Repeat the procedure of figure A-1 for Ts.

c. After the prescribed rated-load drift time, Tdr, repeat procedure of figure A-1 to establish that rated-load governing is stable within requirement during Ts. After (To), change to no-load condition.

d. Find the difference between Cf1 and Cf3. Subtract the small number from the larger. This result is the drift at rated-load (Jr).

e. Repeat 1.5(b.) for no-load governing.

f. Repeat 1.5(c.) for no-load drift time, Tdo, and no-load governing.

g. Repeat 1.5(d.) for Cn2 and Cn3. This result is the drift at no-load, Jo.

1.6 Engine governing drift sample calculations.

a. Assume Cn1 = 61.0 cycles.

b. Assume Cf1 = 60.1 cycles.

c. Assume Cf3 = 59.9 cycles.

d. Jr = Cf1 - Cf3
Jr = 60.1 - 59.9 = 0.2 cycles
Jr = 0.2 divided by 60 = 0.33%

Specification requirement for Jr is a maximum of ± 1.0 percent. The rated-load drift is within specification, since Jr (0.33 percent) is less than the maximum allowed of ± 1.0 percent.

e. Assume Cn2 = 61.2 cycles.

f. Assume Cn3 = 61.4 cycles.

g. Jo = Cn3 - Cn2
Jo = 61.4 - 61.2 = 0.2 cycle
Jo = 0.2 divided by 60 = 0.33%

Specification requirement for Jo is a maximum of ± 1.0 percent. The no-load drift is within specification, since Jo (0.33 percent) is less than the maximum allowed of ± 1.0 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

ANNEX II

ENDURANCE TEST PROCEDURES FOR MODELS
1A08, 2A016, 4A032, 2A042 AND 4A084 MILITARY DESIGN ENGINES

2.0 SCOPE

2.1 Scope. The purpose of these tests is to discover deficiencies in production engines due to design or manufacturing prior to extensive production.

2.2 General conditions. The following general conditions are applicable to all tests.

2.2.1 Instrumentation. Instrumentation shall be in accordance with 4.6.2.2 of the specification and complete test data shall be recorded at 1-hour intervals, except that barometer and vapor pressure and fuel consumption shall be recorded at intervals of not more than 24 hours. Additional instrumentation may be applied at the option of the contracting officer.

2.2.2 Standard atmospheric conditions. All values of observed engine power output at wide-open-throttle during performance tests shall be corrected to standard conditions of 29.00 inches of mercury barometric pressure and 85 °F air temperature by use of the following formula:

$$\text{Corrected BHP} = \text{observed BHP} \times \frac{29.00}{B-e} \times \frac{460+T}{545}$$

Where B = true barometer, in. hg
e = water vapor pressure, in. Hg
T = intake air temperature, °F

2.2.3 Barometric pressure. A mercury barometer of the weather bureau type shall be used to establish the barometric pressure at the engine test site. The barometer reading shall be corrected for temperature and shall be correct to within ±0.05-inch of mercury. An aneroid-type barometer shall not be used.

2.2.4 Vapor pressure. Wet- and dry-bulb temperatures shall be taken for determination of vapor pressure in accordance with U.S. Department of Commerce Publication 235, Weather Bureau Psychrometric Tables. The psychrometer shall be operated near the air intake of the engine in a manner to determine the true relative humidity of the air entering the engine.

2.2.5 Test conditions. All tests shall be conducted under the conditions specified in 4.6.2.1 of the specification as follows:

a. Fuel for the test shall be in accordance with VV-G-1690 regular grade, class A, lead content 4.23 gm/gal maximum, Reid vapor pressure less

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

than 63 KPa. Lubricating oil for the test shall be in accordance with MIL-L-2104, grade 30. A certified chemical analysis shall be required for each new batch of fuel and lubricating oil.

b. An inspection record shall be maintained covering results of inspection on each engine and the maintenance and servicing required both before and during the inspection.

c. No air circulation aids other than the engine-cooling fan shall be directed toward the engine during all tests.

d. An auxiliary fuel system may be connected into the regular fuel system of the engine; however, the fuel supplied to the carburetor shall be by use of the engine fuel pump and not by gravity or other means of external pressurization.

2.3 Tests. Tests shall be performed as follows:

a. In event of any component failure, repeated minor failures, or major engine failure, the test shall be stopped immediately and the failed part shall be inspected in detail for compliance with the requirements of this specification.

b. If an engine fails to perform in accordance with the established requirements, the Government shall stop acceptance of subsequent lots of engines until objective evidence has been provided by the contractor that corrective action has been taken to eliminate the condition which caused the failure.

c. Based on the engineering data, test results or calculations submitted by the manufacturer following failure, the Government may require a partial or complete rerun of this test.

2.3.1 Break-in. The engine(s) shall have a break-in period as follows:

Run	Fraction of rated load	Duration (hours)	Speed (rpm ± 50)
1	0	0.25	2,000
2	0	0.25	2,700
3	0.25	2.0	2,700
4	0.25	4.0	3,600
5	0.75	5.0	3,600
6	1.00	6.0	3,600
7	1.00	0.50	4,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

2.3.2 Initial performance test. Immediately following the "break-in", the carburetor, breaker points (if any), engine timing (if adjustable, adjust if necessary), and the governor shall be set so that it will not interfere with the throttle setting during this test. The maximum corrected horsepower, maximum torque, and fuel consumption shall then be measured over the speed range of the engine (2000 - 4000 rpm), and curves shall be plotted from data for each 400 rpm, descending from 4000 rpm.

2.3.3 Governor test. After the completion of the initial performance test (see 2.3.2), a governor test in accordance with 4.7.6 of the specification shall be made, except that Ts shall be 2 hours for models 1A08 and 2A016 and 4 hours for models 4A032, 2A042 and 4A084. Test schedule shall be 2 hours at both no-load and rated-load for models 1A08 and 2A016 and 4 hours at both no-load and rated-load for models 4A032, 2A042 and 4A084.

2.3.4 Endurance tests. Following completion of the initial performance tests (see 2.3.2) and the governor test (see 2.3.3), endurance tests in accordance with 4.7.8 of the specification shall be run as follows:

a. The rated-load test shall be in accordance with 4.7.4 of the specification except that the operating time shall be 1,500 hours. At the completion of each 500-hour period, the governor test (see 2.3.3) shall be run again.

b. The wide-open-throttle test shall be run for 500 hours with speed maintained between 3,550 and 3,650 rpm. Horsepower load shall be not less than 2.1 horsepower for model 1A08, 4.25 horsepower for model 2A016, 9.0 horsepower for model 4A032, 15 horsepower for model 2A042, and 29.75 horsepower for model 4A084 engines.

2.3.5 Performance test at completion of endurance tests. Immediately following the completion of endurance tests, the engine shall undergo another performance test (see 2.3.2). The results of this test shall be plotted. Only a minor adjustment to the carburetor shall be permitted.

2.3.6 Final performance test. After completion of the performance test (see 2.3.5), clean valves and combustion chambers, install new spark plugs, and adjust the carburetor, breaker points (if any), and timing (if adjustable) to specified conditions. A final performance test shall be run and shall be plotted.

2.4 Servicing and adjustment schedule. Servicing and adjustment shall be recorded after performance in accordance with table A-I of this appendix.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

2.5 Final approval and acceptance. Final approval and acceptance of the endurance engines by the Government shall be in accordance with 4.7.7 of the specification. The endurance test engines shall meet or exceed the following requirements:

a. The average specific oil consumption shall not exceed 0.006 lb/bhp/hr.

b. Servicing, adjustments and maintenance in excess of that allowed by 2.4 shall not be permitted.

c. The model 1A08 engine shall develop not less than 1.5 bhp; model 2A016, 3.00 bhp; model 4A032, 6.00 bhp; model 2A042, 10 bhp and model 4A084, 20 bhp with speed maintained at a nominal 3,600 rpm (between 3,590 and 3,610 rpm) with ±18 rpm for models 4A032, 2A042 and 4A084, and ±36 rpm for models 1A08 and 2A016 at a stabilized adjusted speed for the duration of the rated-load endurance test. Model 1A08 shall develop not less than 2.1 bhp; model 2A016, 4.25 bhp; model 4A032, 9.0 bhp; model 2A042, 15.0 bhp, and model 4A084, 29.75 bhp during the wide-open-throttle endurance test run, with servicing, adjustments and replacements not exceeding that specified in table A-I of this appendix.

d. Engines shall meet or exceed the governor test requirements (see 2.3.3).

e. After 50 hours of operation, engines shall develop not less than the wide-open-throttle horsepower specified in 3.7 of the specification with speed maintained between 3,588 and 3,612 rpm during the performance tests required by 2.3.2, 2.3.5, and 2.3.6.

f. Lubricating oil temperature shall be not less than 130 °F or more than 250 °F for models 1A08 and 2A016 and not more than 265 °F for models 4A032, 2A042 and 4A084 during all tests. Spark plug head temperature shall not exceed 475 °F at rated-load and 500 °F at wide-open-throttle.

2.6 Final examination. Upon completion of 2.3.6, the engine shall be disassembled and examined in accordance with 4.7.7.

2.7 Final report. A final report for each engine shall be required and shall include all data recorded during the tests, including the curves, photographs and measurement of parts and shall be submitted to Commander, USA Belvoir Research, Development, and Engineering Center, Fort Belvoir, VA 22060-5606, ATTN: STRBE-TS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

TABLE A-I. Servicing and adjustment schedule.

Before test	Hours between service periods during test	Item
X	8	Any minor maintenance such as retightening and retorquing nuts, bolts and screws that can be performed, without disturbing or removing adjacent components, accessories, or shrouding (except top shroud, pump, intake manifold, and exhaust system).
X	None allowed	Parts, components, accessories (except as noted herein), or any major maintenance that requires removal of other components, accessories or shrouding, such as oil pans, cylinder heads and connecting rods.
X	*	Cleaning, regapping or replacing spark plugs.
X	*	Servicing fuel filter.
X	-	Checking compression pressures.
X	8	Adding lubricating oil.
X	8	Adjusting carburetor.
X	*	Cleaning carburetor.
X	8	Adjusting governor (normal speed adjustment).
X	100	Changing lubricating oil.
X	100	Changing lubricating oil filter element (models 4A032, 2A042 and 4A084).
X	100	Servicing air cleaner.
X	*	Adjusting and dressing breaker points (if any).
X	300	Adjusting ignition timing (if adjustable).
-	300	Cleaning combustion chamber and manifolds.
X	250	Replacing breaker points, if any (model 1A08).
-	250	Replacing condenser (if present) or coil.
-	500	Retorque head nuts (models 1A08, 2A016 and 4A032).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

TABLE A-I. Servicing and adjustment schedule - Continued.

Before test	Hours between service periods during test	Item
-	500	Replacing governor rod housing (models 1A08 and 2A016).
X	Not allowed, except where component is removed for servicing.	Gasket replacement.
-	500	Adjust valve lash (intake and exhaust).

* As required.
X Required.
- Allowed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

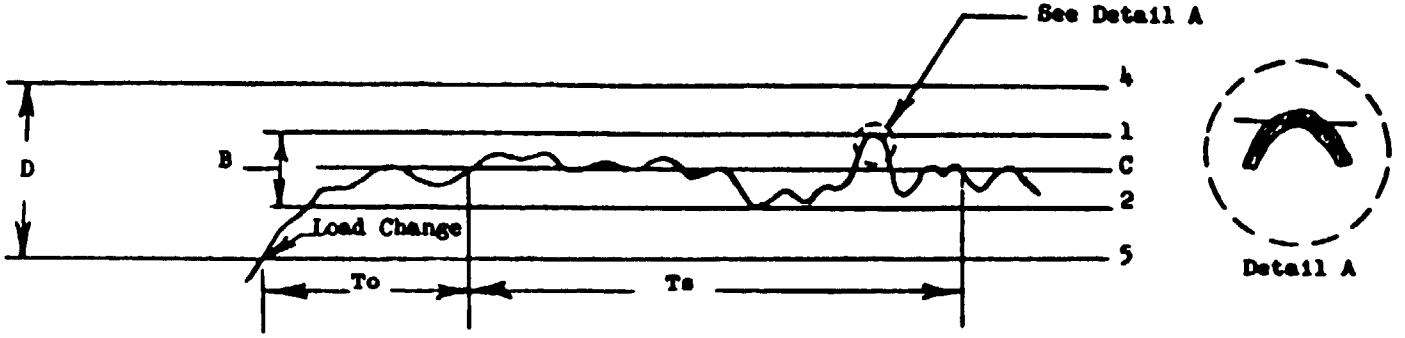
FAILURE DEFINITION

A chargeable reliability failure shall be defined as an event, other than scheduled maintenance, in which malfunction occurs, resulting in: (1) the replacement of the malfunctioning item (part, component, subassembly, or assembly) by a like item, except as provided in table A-I; or (2) the removal and repair of the malfunctioning item or any repair required because of faulty workmanship (as defined explicitly or quantitatively in the Technical Data Package (TDP) in manufacture or assembly.

Additionally, any impending malfunction detected which constitutes a safety hazard to operating personnel, or which would cause serious damage to the equipment if continued in operation, shall be considered a chargeable failure.

Those malfunctions occurring as dependent or secondary failures or as a result of improper maintenance procedures or operator error shall be excluded from consideration as chargeable failures. Those malfunctions resulting in a proven design modification applicable to all production items, shall also satisfy both the test agency and the test sponsor as to its effectiveness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.



Procedure.

A Draw line 1 parallel to Axis of chart movement intersecting center of trace width at maximum excursion of trace during test time sample Ts.

B. Draw line 2 in the same manner as Step A for minimum excursion.

C. Establish values of line 1 and 2 from actual chart, take average and draw obtained value as Line C.

D. Multiply stability specification percentage requirement by rated frequency or speed. Add this value to C to obtain line 4 and subtract from C to obtain Line 5. Lines 4 and 5 are the specification limits.

E. If the value of Line 1 is less than Line 4 and the value of Line 2 is greater than Line 5 during test sample Ts the governing is stable within specifications.

NOTES:

(1) All charts are to be read from right to left with speed increasing from bottom to top of page.
(2) Test time sample Ts should be at least 2 seconds (To) from any load change.
(3) The load change will be considered to occur when the trace passes out of the specified speedband.
(4) To determine chart length for time Ts multiply chart speed (In/Min) X required time (Min).

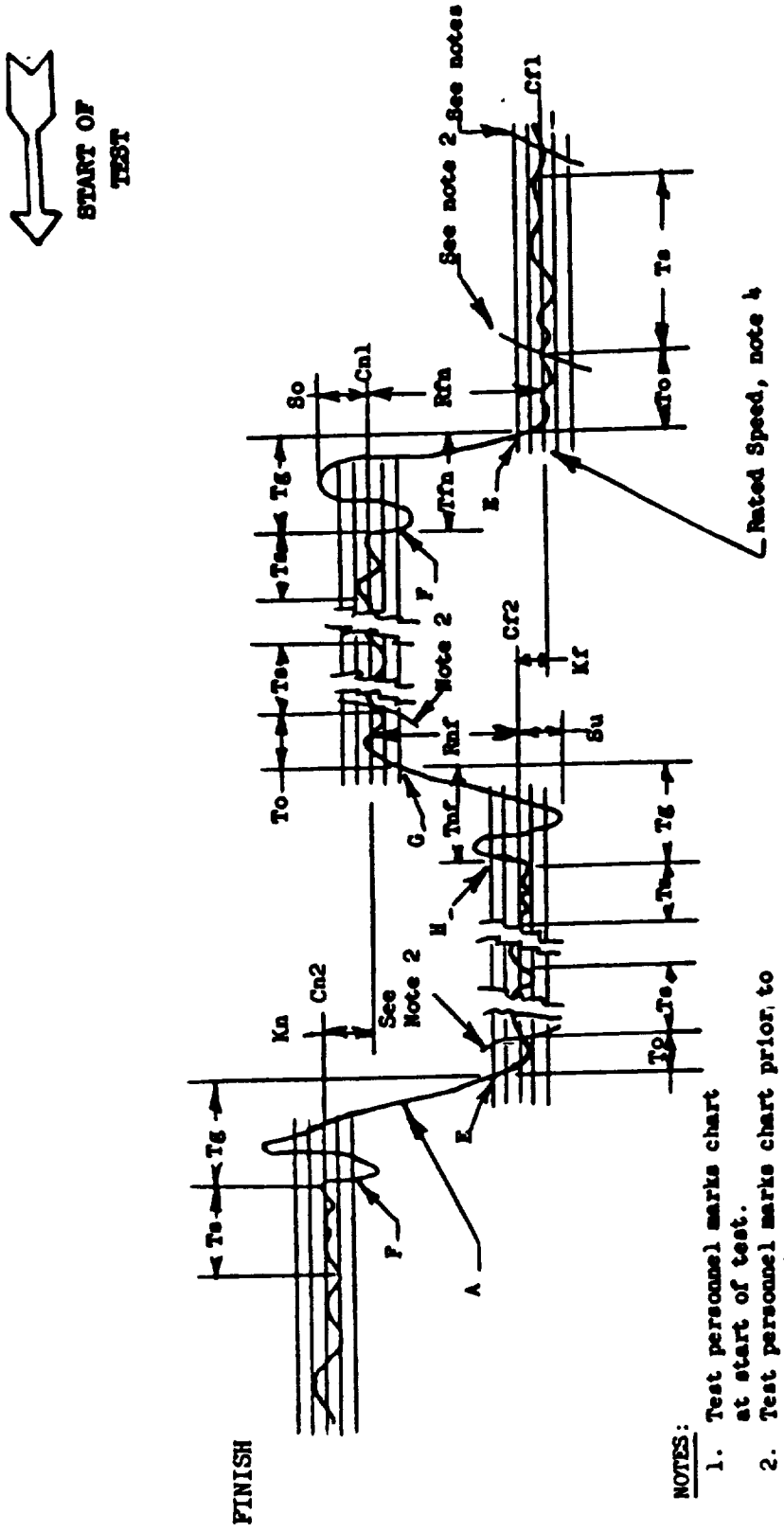
EXAMPLE: (Assumed values of frequency and Specification)

1. Line 1 = 59.4
2. Line 2 = 59.1
3. Line C = (59.4 / 59.1) ÷ 2 = 59.25
4. Stability from specification = ∠ 1/2 of 1% = ∠ (.01 X 60) ÷ 2 = ∠ 0.3 cycles
5. Line 4 = Line C ∠ 0.3
   = 59.25 / 0.3 = 59.55
6. Line 5 = Line C - 0.3
   = 59.25 - 0.3 = 58.95

This governor is stable since Line 4 is greater than Line 1 and Line 5 is less than Line 2.

FIGURE A-1. Engine governing stability trace analysis. X828A

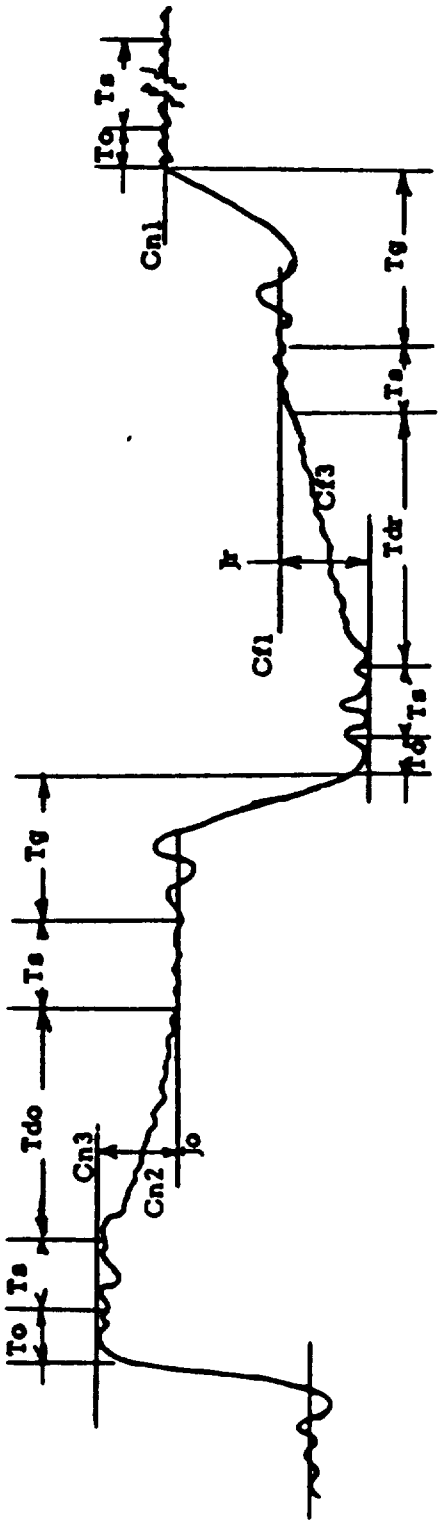
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.



NOTES:
1. Test personnel marks chart at start of test.
2. Test personnel marks chart prior to making a load change.
3. Read from right to left.
4. Rated Speed does not necessarily coincide with Cf1.

FIGURE A-2. Speed stability, transient response and hysteresis analysis. X829

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.



NOTES:

1. The Tdr and Tdo times are normally several hours whereas Ts and To are less than a minute.
2. Any apparent drift occurring during test time samples Ts is to be disregarded and the analysis performed as previously in accordance with Figure A-1.

FIGURE A-3. Engine governor drift analysis.

X830

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-46717G	Engines, Gasoline: Air-Cooled, Industrial-Type 4-Cycle,

3a. NAME OF SUBMITTING ORGANIZATION Military-Design

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

b. ADDRESS *(Street, City, State, ZIP Code)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
USA Belvoir RDE Center
STRBE-TSE
Fort Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:46Z
Check the source to verify that this is the current version before use.